VERSION 1 — FROZEN METHODOLOGY

Project Title

Deterministic Monocular Spatial Reasoning System (Version 1)

Status

FROZEN — This document defines the baseline methodology. No calibration-by-motion, self-height recovery, or sensor-assisted scale injection is included in this version.

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1. Core Objective

To compute **object distance, direction, and height** using a **single monocular camera**, operating **fully offline**, without LiDAR, cloud services, object-size priors, or neural depth hallucination.

The system prioritizes **deterministic geometry**, explainability, and graceful failure over instantaneous precision.

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2. Fundamental Assumptions

Version 1 assumes the following are available or pre-known:

- Fixed camera intrinsics (focal length, sensor size, field of view)
- Known camera height above the ground (externally provided or preconfigured)
- Ability to infer gravity-aligned orientation (via scene geometry)
- Presence of a detectable ground plane or valid proxy surface

No assumption is made about:

- Object dimensions
- Object class typical sizes
- Internet connectivity
- LiDAR or depth sensors

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3. Primary Methodologies

3.1 Ground-Plane Distance Estimation

Principle:
Any object that touches the ground can have its distance computed by identifying the ground contact point and measuring the viewing angle from the camera to that point.

Invariant Used:
Camera height relative to ground + angular geometry.

This provides absolute distance without knowing object size.

3.2 Virtual Vertical Line Projection (Suspended Objects)

Principle:
For objects not in contact with the ground (e.g., signs, shelves), a virtual vertical line is dropped from the object's visible base until it intersects the ground plane.

Distance is then computed at the intersection point using the same ground-plane logic.

This extends ground-based measurement to suspended hazards.

3.3 Virtual Ground / Elevated Reference Plane

Principle:
When objects rest on elevated surfaces (tables, counters), the system redefines the ground plane virtually by elevating it to the detected support surface height.

All distance calculations reuse the same geometry, eliminating special-case logic.

This unifies floor-level, table-level, and suspended-object handling.

3.4 Horizon-Based Height Estimation

Principle:
A gravity-aligned, perfectly horizontal reference line originating from the camera defines a zero-elevation baseline.

Object height is computed by measuring angular displacement above or below this reference line and combining it with ground-plane distance.

Height recovery is deterministic and geometry-driven.

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4. Spatial Outputs

Version 1 produces:

- Distance to object (meters)
- Horizontal direction relative to camera (degrees)
- Vertical position (above/below camera)
- Height above ground (if camera height is known)

All outputs are accompanied by confidence bounds derived from geometric consistency.

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5. Explicit Exclusions (Not in Version 1)

The following are **intentionally excluded**:

- Camera self-height estimation
- Motion-based scale recovery
- Accelerometer-based distance integration
- Visual–inertial odometry
- Neural monocular depth models
- Fiducial markers
- User interaction during measurement

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6. Design Philosophy

- Deterministic over probabilistic
- Geometry over data
- Offline-first
- Explainable failure modes

- Minimal assumptions

Version 1 serves as the **reference baseline** against which all future extensions must be justified.

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7. Version Lock

This document defines **Version 1**.

Any future additions (e.g., calibration rituals, motion-based height recovery, sensor fusion) must be specified as **Version 2+ extensions** and must not retroactively alter the assumptions or guarantees of Version 1.

END OF VERSION 1